21ASC305T
ROCKET PROPULSION

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- ***Unit-1 - Basic Concepts 9 Hour***
- *Rocket propulsion systems -working principle, Classification of rockets, Rocket equation, Rocket Nozzles and its classifications -Nozzle performance- equilibrium and frozen flow, Characteristic velocity and Thrust coefficient, Performance parameters and efficiencies of rocket, Staging and clustering of rocket*
- ***Unit-2 - Chemical Propellants 9 Hour***
- *Chemical propellants -Molecular mass, Specific heat capacities, Specific heat ratio, mixture ratio, Stoichiometric ratio and equivalence ratio, Energy release during combustion-Heat of formation and Heat of combustion-Combustion Instabilities-Criterion for choice of propellant, Solid Propellants-Composition and processing, Liquid Propellants-Classifications of liquid propellant, Storability of liquid propellant, Chemical Equilibrium*
- ***Unit-3 - Solid Rocket Engine 9 Hour***
- *Propulsion Elements for Solid Rocket Motors, Solid propellant combustion and Internal Ballistics of Motors - Mechanism of burning, Ignition System-Igniter types - Solid propellant grain design - Burn Rate-Factors influencing burn rates, burn rate index for stable operation, Action time and burn time, Design of Solid Propellant rocket, Simulation of solid rocket motor performance.*
- ***Unit-4 - Liquid Rocket Engine 9 Hour***
- *Liquid Propellant Rocket-Hardware components and its functions, Propellant feed systems –Gas pressure feed system, Turbo pump feed systems and engine cycles, Tank Pressurization - Types of injectors – Thrust chamber, Cooling of Thrust chamber - Cryogenic propulsion system and its special features, Simulation of Liquid rocket performance, Hybrid rockets- Review of solid-fuel regression rate behavior in classical and non-classical Hybrid Rocket Motors - Rocket engine design using software.*
- ***Unit-5 - Non-conventional Propulsion Techniques 9 Hour***
- *Non - conventional propulsion techniques -Electrical Rockets-Electro-thermal propulsion system-Arc-jet thruster, Resistojet Thruster-Electro-static thruster-Ion thruster, Bombardment Ionization, Hall Effect Thruster -Electro-magnetic propulsion system-Magneto plasma dynamic accelerator, Pulsed Plasma accelerator- Laser Propulsion -Nuclear powered rockets- Solar Propulsion system*

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- *1. George P. Sutton, Oscar Biblarz, "Rocket propulsion elements", Wiley India Pvt Ltd. Ninth Edition, 2017*
- *2. Ramamurthi.K, "Rocket propulsion", Laxmi Publications, India, Second edition, 2019.*
- *3. J D. Mattingly and K M. Boyer,"Elements of Propulsion: Gas Turbines and Rockets", Second Edition,2016*
- *4. Philip Hill and Carl Peterson, "Mechanics and thermodynamics of propulsion", Pearson India, second edition, 2010.*
- *5. Stephen R. Turns, "An Introduction to Combustion: Concepts", McGraw-Hill Education; 4th edition, 2020*
- *6. Yahya S M, "Gas Tables",8th Edition, 2018*

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PROPELLANT FEED SYSTEMS

- Propellant feed systems have two principal functions:
- (1) to **raise the pressure** of the propellants and
- (2) to **supply them at design mass flow rates** to one or more thrust chambers.

- The energy for these functions comes either from a high-pressure gas, centrifugal pumps, or a combination of the two.

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PROPELLANT FEED SYSTEMS

- The selection of a particular feed system and its components is governed primarily by the
  - rocket application,
  - duration,
  - number or type of thrust chambers,
  - past experience,
  - mission and
  - by the general requirements of simplicity of design,
  - ease of manufacture, low cost, and minimum inert mass.

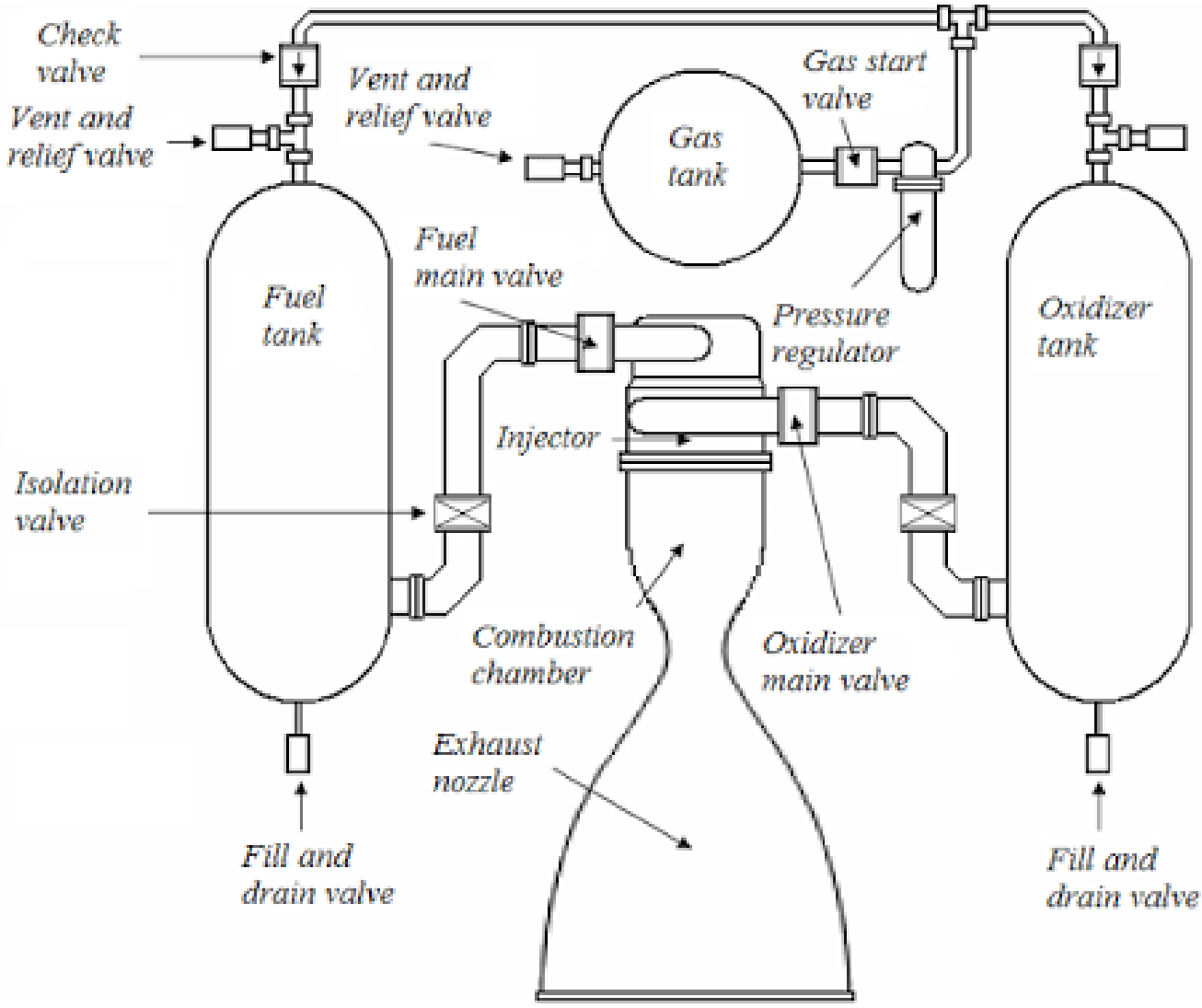
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PROPELLANT FEED SYSTEMS

- In general, *gas pressure feed systems* give superior performance to turbopump systems when the
  - vehicle's total impulse or propellant mass is relatively low,
  - the chamber pressure is relatively low,
  - the engine thrust-to-weight ratio is low (usually less than 0.6)

- heavy-walled propellant tanks and the pressurizing gas constitute the major inert mass of the engine system

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Check valve
Vent and relief valve
Vent and relief valve
Gas tank
Gas start valve
Fuel main valve
Fuel tank
Pressure regulator
Oxidizer tank
Injector
Isolation valve
Combustion chamber
Oxidizer main valve
Exhaust nozzle
Fill and drain valve
Fill and drain valve

PROPELLANT FEED SYSTEMS

- In a *turbopump feed system*, propellant tank pressures are much lower (by a factor of 10 to 40) and thus vehicles' tank masses are much lower (by the same factor).

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GAS PRESSURE FEED SYSTEMS

- One of the simplest and most common means of pressurizing liquid propellants and force them out of their respective tanks is to use a high-pressure gas.

- There are **two common types** of pressurized feed systems both still used often today.

- The first uses a ***gas pressure regulator*** in the gas feed line with the engine operating at essentially constant tank pressure and nearly constant thrust.

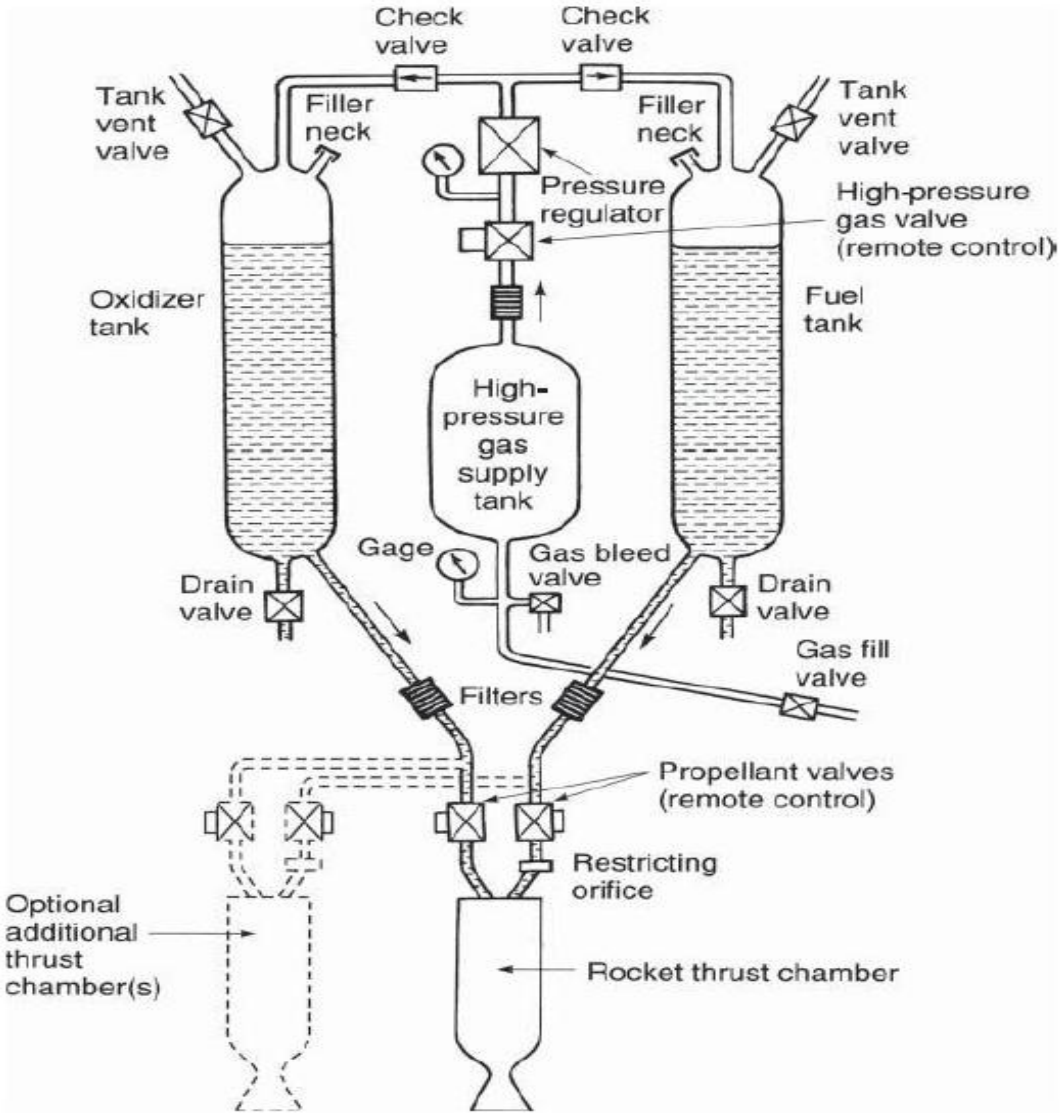
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GAS PRESSURE FEED SYSTEMS

- A high-pressure gas tank,
- a gas starting valve,
- a gas pressure regulator,
- propellant tanks,
- propellant valves, and
- feed lines.

- Additional components, such as
  - filling and draining provisions,
  - check valves,
  - filters,
  - flexible elastic bladders for separating the liquid from the pressurizing gas, and
  - pressure sensors or gauges,

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Check valve
Check valve
Tank vent valve
Filler neck
Filler neck
Tank vent valve
Pressure regulator
High-pressure gas valve (remote control)
Oxidizer tank
Fuel tank
High-pressure gas supply tank
Gage
Gas bleed valve
Drain valve
Drain valve
Gas fill valve
Filters
Propellant valves (remote control)
Restricting orifice
Optional additional thrust chamber(s)
Rocket thrust chamber

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GAS PRESSURE FEED SYSTEMS

- When the propellants are completely consumed, the pressurizing gas can also be used to scavenge and clean lines and valves of much of the liquid propellant residue.

- If a unit is to be **used and flown repeatedly**, such as a space-maneuver rocket, it may include several additional features such as a thrust-regulating device and a tank level gauge

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GAS PRESSURE FEED SYSTEMS

- The second common type of gas pressure feed system is called a ***blow-down feed system***.

- Propellant tanks are larger because they store not only the propellants but also the pressurizing gas at an initial maximum propellant tank pressure.

- There is no separate high-pressure gas tank and no pressure regulator.

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GAS PRESSURE FEED SYSTEMS

- The expansion of the gas already in the tanks provides for the expulsion of the propellants.

- Monopropellants gas pressure feed systems are simpler since there is only one propellant, thus reducing the number of pipes, valves, and tanks.

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GAS PRESSURE FEED SYSTEMS

Type	Regulated Pressure	Blowdown
Pressure/thrust	Stays essentially constant[a]	Decreases as propellant is consumed
Gas storage	In separate high-pressure tanks	Gas is stored inside propellant at tank pressure with large ullage volume (30–60%)
Required components	Needs regulator, filter, gas valves, and gas tank	Larger, heavier propellant tanks
Advantages	Nearly constant-pressure feed gives essentially constant propellant flow and approximately constant thrust, I_s and r	Simpler system. Less gas required. Can be less inert mass.
	Better control of mixture ratio	No high-pressure gas tank
Disadvantages	Slightly more complex	Thrust decreases with burn duration
	Regulator introduces a small pressure drop	Somewhat higher residual propellant due to less accurate mixture ratio control
	Gas stored under high pressure, often for a long time	Thruster must operate and be stable over wide range of thrust values and modest range of mixture ratios
	Requires more pressurizing gas	Propellants stored under pressure; slightly lower I_s toward end of burning time

- There are **several *sources of pressurizing gas*** used in tank pressurization systems.

- 1. ***High-pressure inert gases*** stored at ambient temperature are the most common.
- Typical gases are helium, nitrogen, and air.

- 2. ***Heated high-pressure inert gases*** (typically 200 to 800 ◦F or 93 to 427 ◦C) reduce the amount of required gas and thus the inert mass of the pressurizing system.

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- ***3. Gases created by a chemical reaction*** using either liquid bipropellants or a monopropellant, or alternatively a solid propellant, all at mixture ratios or compositions that result in "warm gas."

- **Chemically generated warm gases** usually result in tank pressurization systems lighter than heated inert gas systems, particularly for high total impulse applications.

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- 4. ***Evaporated flow of a small portion of a cryogenic liquid propellant***, usually liquid hydrogen or liquid oxygen, by applying heat from a thrust chamber cooling jacket or from turbine exhaust gases (with heat exchangers), and then using a part or all of this evaporated flow for tank pressurization.

- 5. ***Direct injection* of a small stream** of hypergolic fuel into the main oxidizer tank and a small flow of hypergolic oxidizer into the fuel tank has been tried by several countries but with limited success.

- 6. ***Self-pressurization of cryogenic propellants*** by evaporation is feasible but can be difficult to control.

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- In order to **design or analyze** any pressurizing system it is necessary to have relevant information about the tank and the engine.

- This can include basic engine parameters, such as
  - propellant flow,
  - thrust,
  - duration,
  - pulse width,
  - propellant tank volume,
  - percent ullage of tank volume,

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- percent ullage of tank volume,
- storage temperature range,
- propellant and pressurizing gas properties,
- propellant tank pressure, gas tank pressure, and/or
- amount of unavailable residual propellant.

- For many of these items, nominal, maximum, and minimum values may be needed.

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